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RBE211TC DYNAMIC SYSTEMS

VIBRATION OF MECHANICAL SYSTEMS – MDOF SYSTEMS (PART 2)

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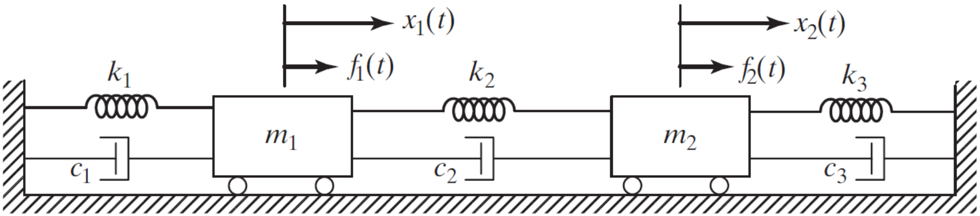
Lecture Outline

1. Recap of MDOF matrix equations
2. Undamped free vibration assumption
3. Harmonic solution assumption
4. Eigenvalue problem
5. Natural frequencies
6. Mode shapes
7. Physical interpretation of each mode
8. Worked 2DOF example

Recap: Matrix Form of an MDOF System



General matrix form:

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{f(t)\}$$

where:

- $[M]$: mass matrix
- $[C]$: damping matrix
- $[K]$: stiffness matrix
- $\{x\}$: displacement vector
- $\{f(t)\}$: force vector

Undamped Free Vibration

For free vibration:

$$\{f(t)\} = \{0\}$$

If damping is neglected:

$$[C] = [0]$$

Then the equations reduce to

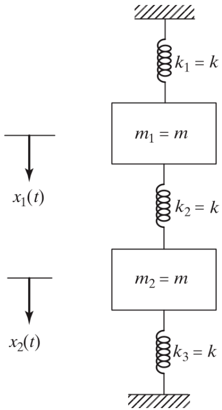
$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

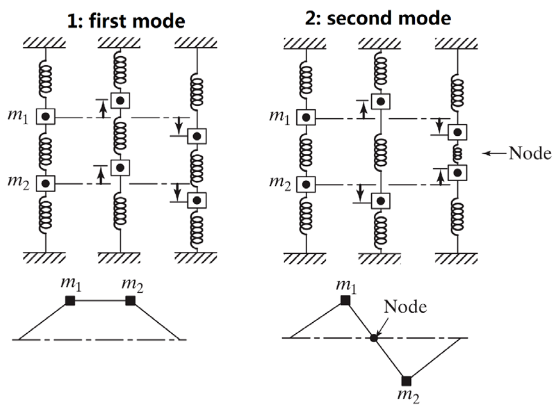
Why Does an MDOF System Have Multiple Natural Frequencies?

An MDOF system has:

- more than one independent coordinate
- more than one possible vibration pattern
- more than one natural frequency

Key idea:
Each characteristic vibration pattern corresponds to a mode.





Assume Harmonic Motion

Assume the free-vibration response takes the form

$$\{x(t)\} = \{X\}\sin(\omega t + \phi)$$

or equivalently

$$\{x(t)\} = \{X\}e^{j\omega t}$$

where:

- $\{X\}$: amplitude vector
- ω: natural frequency
- ϕ: phase angle

Substitute the Harmonic Solution

For

$$\{x(t)\} = \{X\}\sin(\omega t + \phi)$$

we have

$$\{\ddot{x}(t)\} = -\omega^2\{X\}\sin(\omega t + \phi)$$

Substitute into

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

to obtain

$$\left([K] - \omega^2[M]\right)\{X\} = \{0\}$$

The Eigenvalue Problem

The equation

$$([K] - \omega^2 \, [M])\{X\} = \{0\}$$

has a nontrivial solution only if

$$\det\left([K] - \omega^2 [M]\right) = 0$$

This is called the **characteristic equation** or **frequency equation**.

Natural Frequencies and Mode Shapes

From

$$\det\big([K] - \omega^2 [M]\big) = 0$$

we obtain:

- natural frequencies: $\omega_1,\ \omega_2, \cdots$
- mode shapes:$\{X\}^{(1)}, \{X\}^{(2)}, \cdots$

Key interpretation:
Each natural frequency is associated with one characteristic shape of motion.

What Does a Mode Shape Mean Physically?

A mode shape tells us:

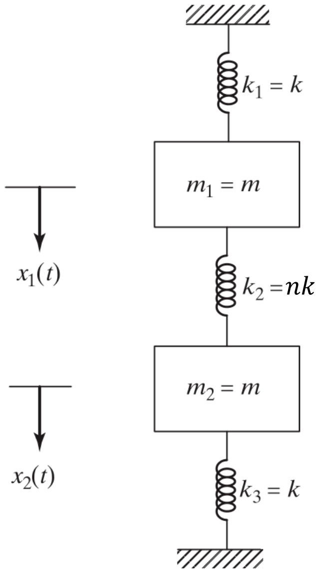
- the relative motion of the coordinates
- whether the coordinates move in phase or out of phase
- the amplitude ratio between coordinates

Key point:
Mode shapes describe the pattern of vibration, not the absolute size of motion.

Example: Simple Two-Degree-of-Freedom System

Consider the undamped 2DOF system:

- two equal masses m
- outer springs of stiffness k
- middle spring of stiffness nk



Coordinates:

- $x_1(t), x_2(t)$

Find:

- natural frequencies
- mode shapes

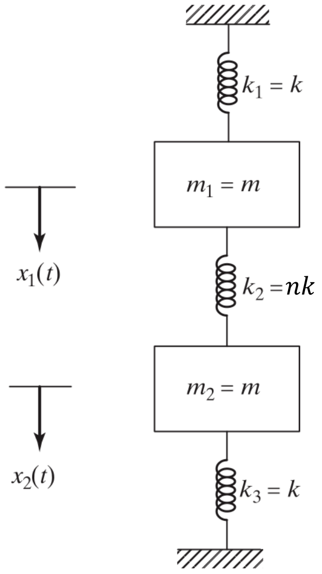
Example: Simple Two-Degree-of-Freedom System

The equations of motion are

$$m\ddot{x}_1 + (k + nk)x_1 - nkx_2 = 0$$
$$m\ddot{x}_2 + (k + nk)x_2 - nkx_1 = 0$$

or

$$m\ddot{x}_1 + (1 + n)kx_1 - nkx_2 = 0$$
$$m\ddot{x}_2 + (1 + n)kx_2 - nkx_1 = 0$$



Example: Simple Two-Degree-of-Freedom System

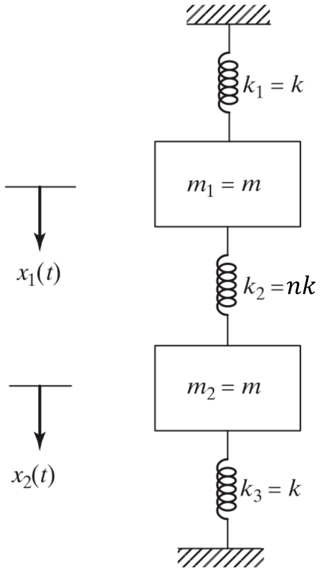
The equations can be written as

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

with

$$[M] = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}$$

$$[K] = \begin{bmatrix} (1+n)k & -nk \\ -nk & (1+n)k \end{bmatrix}$$



Example: Simple Two-Degree-of-Freedom System

Assume

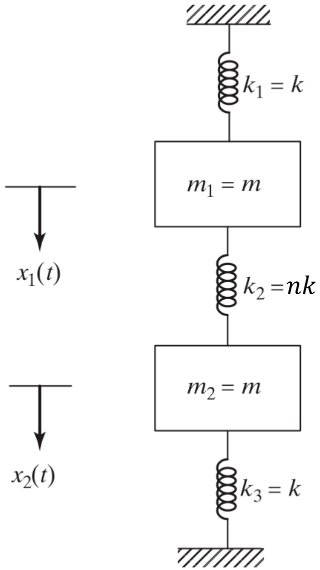
$$\{x(t)\} = \{X\}\sin(\omega t + \phi)$$

where

$$\{X\} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}$$

Then

$$\big([K] - \omega^2[M]\big)\{X\} = \{0\}$$



Example: Simple Two-Degree-of-Freedom System

For a nontrivial solution,

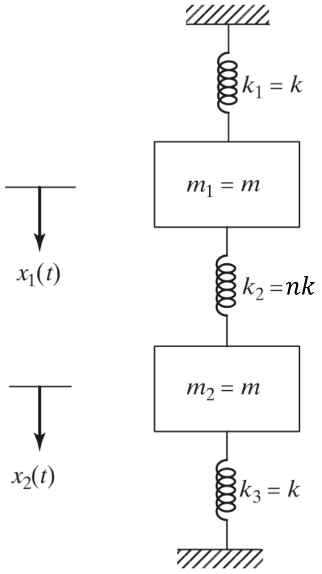
$$\det\left([K] - \omega^2 [M]\right) = 0$$

So

$$\begin{vmatrix} (1+n)k - m\omega^2 & -nk \\ -nk & (1+n)k - m\omega^2 \end{vmatrix}$$

Hence

$$\left((1+n)k - m\omega^2\right)^2 - (-nk)^2 = 0$$



Example: Simple Two-Degree-of-Freedom System

From

$$\left((1+n)k - m\omega^2\right)^2 - (-nk)^2 = 0$$

we get

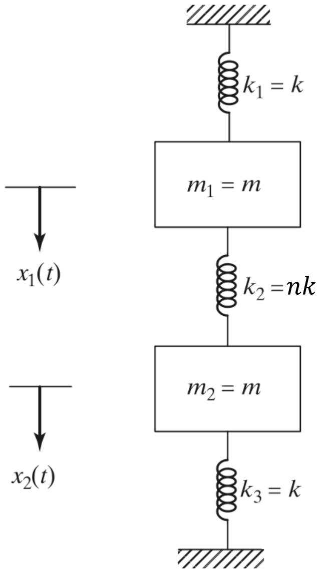
$$(1+n)k - m\omega^2 = \pm nk$$

Thus

$$\omega_1^2 = \frac{k}{m}, \qquad \omega_2^2 = \frac{(1+2n)k}{m}$$

So

$$\omega_1 = \sqrt{\frac{k}{m}}, \qquad \omega_2 = \sqrt{\frac{(1+2n)k}{m}}$$



Example: Simple Two-Degree-of-Freedom System

For

$$\omega_1^2 = \frac{k}{m}$$

substitute into

$$\left([K] - \omega^2[M]\right)\{X\} = \{0\}$$

to get

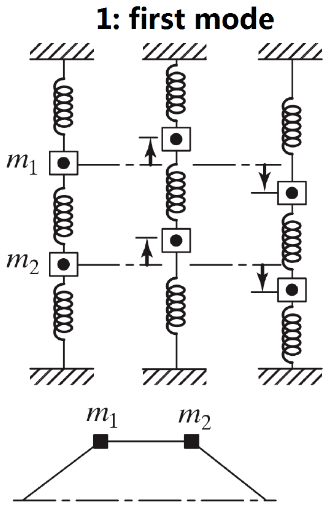
$$\begin{bmatrix} nk & -nk \\ -nk & nk \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Hence

$$X_2 = r_1 X_1, \text{ where } r_1 = 1$$

So the first mode shape is

$$\{X\}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_1^{(1)} \end{Bmatrix}$$



Example: Simple Two-Degree-of-Freedom System

For

$$\omega_2^2 = \frac{(1+2n)k}{m}$$

substitute into

$$\left([K] - \omega^2[M]\right)\{X\} = \{0\}$$

to get

$$\begin{bmatrix} -nk & -nk \\ -nk & -nk \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

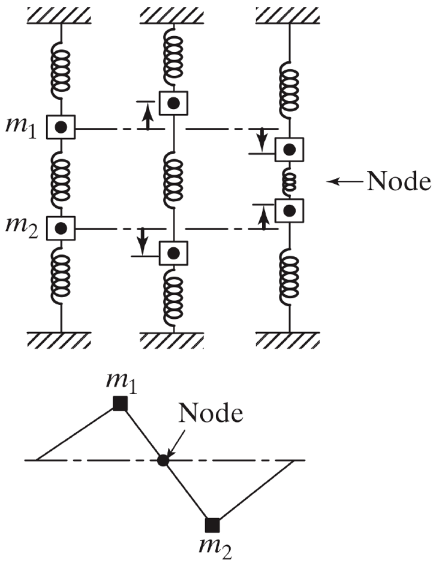
Hence

$$X_2 = r_2 X_1, \text{ where } r_2 = -1$$

So the second mode shape is

$$\{X\}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ -X_1^{(2)} \end{Bmatrix}$$

2: second mode



Physical Interpretation of the Two Modes

First mode:

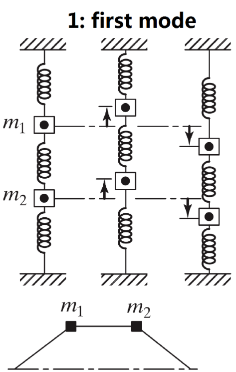
$$\{X\}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_1^{(1)} \end{Bmatrix}$$

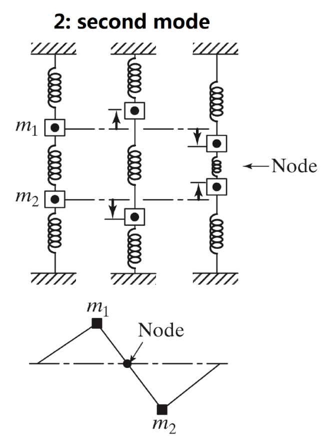
- masses move in phase
- middle spring does not deform significantly

Second mode:

$$\{X\}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ -X_1^{(2)} \end{Bmatrix}$$

- masses move out of phase
- middle spring deforms strongly

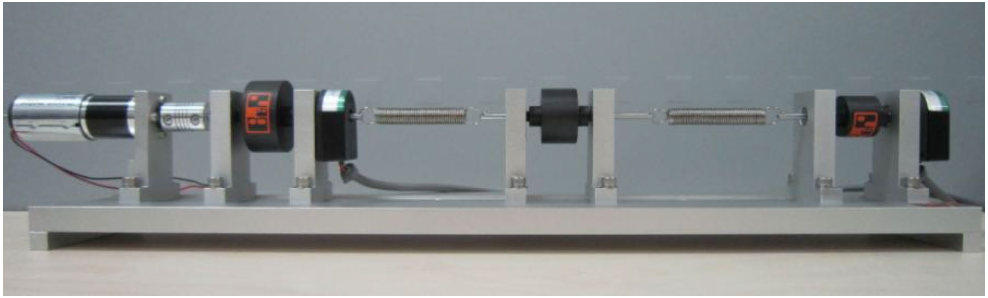




Why Are Mode Shapes Important?

Mode shapes help us:

- visualize how the system vibrates
- identify in-phase and out-of-phase motion
- understand why different natural frequencies exist
- interpret experimental observations



Multi-DOF Rotary Torsional System

Free-Vibration Response as a Combination of Modes

For a 2DOF undamped system, the total response can be written as

$$\{x(t)\} = \{X\}^{(1)}(A_1 \cos \omega_1 t + B_1 \sin \omega_1 t) + \{X\}^{(2)}(A_2 \cos \omega_2 t + B_2 \sin \omega_2 t)$$

Key idea:
The total motion is a superposition of modal contributions.

How Do Initial Conditions Affect the Motion?

Initial conditions determine:

- how much of each mode is excited
- whether one mode dominates
- whether both modes appear in the response

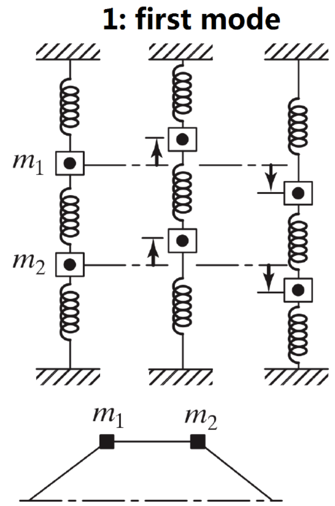
Examples:

- symmetric initial displacement may excite mainly the first mode
- antisymmetric initial displacement may excite mainly the second mode

Example: Initial Conditions That Excite Mainly the First Mode

If the initial displacement is proportional to

$$\{X\}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_1^{(1)} \end{Bmatrix}$$



then the system tends to vibrate mainly in the first mode.

Interpretation:

- coordinates move together
- in-phase motion dominates

Example: Initial Conditions That Excite Mainly the Second Mode

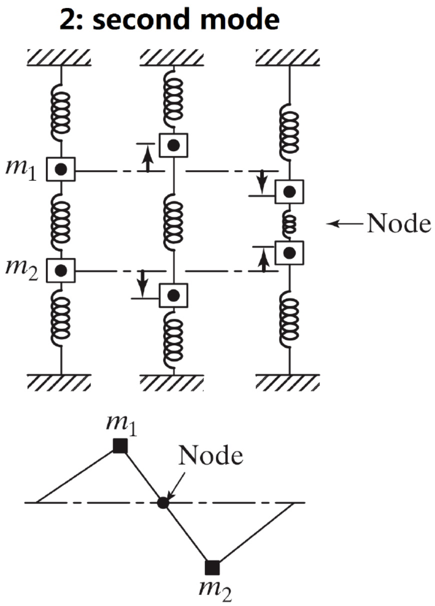
If the initial displacement is proportional to

$$\{X\}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ -X_1^{(2)} \end{Bmatrix}$$

then the system tends to vibrate mainly in the second mode.

Interpretation:

- coordinates move oppositely
- out-of-phase motion dominates

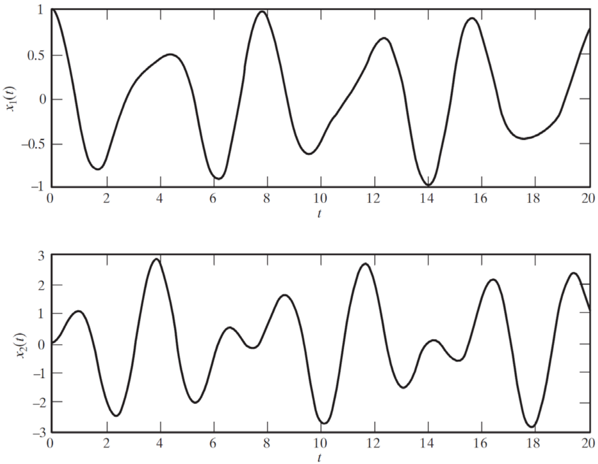


When Multiple Modes Are Excited

In general:

- arbitrary initial conditions excite more than one mode
- the total response contains multiple frequencies
- the motion may appear more complicated than SDOF vibration

Key point:
Multi-frequency response is natural in MDOF systems.

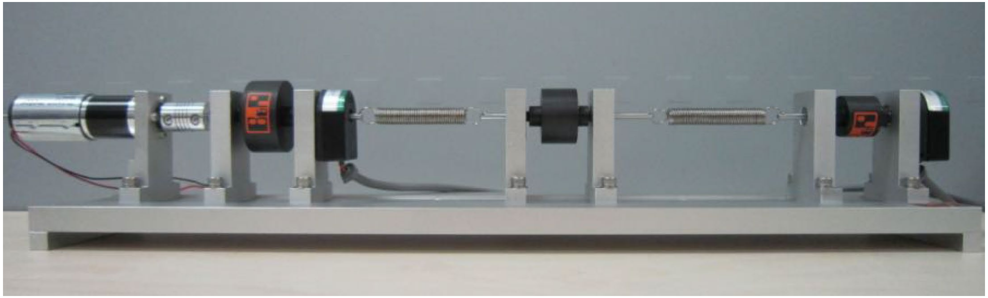


Natural response of an undamped TDOF system.

Connection to Lab 5

Lab 5 will help illustrate:

- multiple natural frequencies
- coupled motion
- mode-dependent response
- comparison between theory, simulation, and experiment



Multi-DOF Rotary Torsional System

Review

- Undamped free vibration of MDOF systems
- Eigenvalue problem
- Natural frequencies
- Mode shapes
- Modal interpretation
- Modal superposition
- Link to experiment